

Machine Learning Interatomic Potentials for Exploring Surface Features of Bulk Metallic Glasses

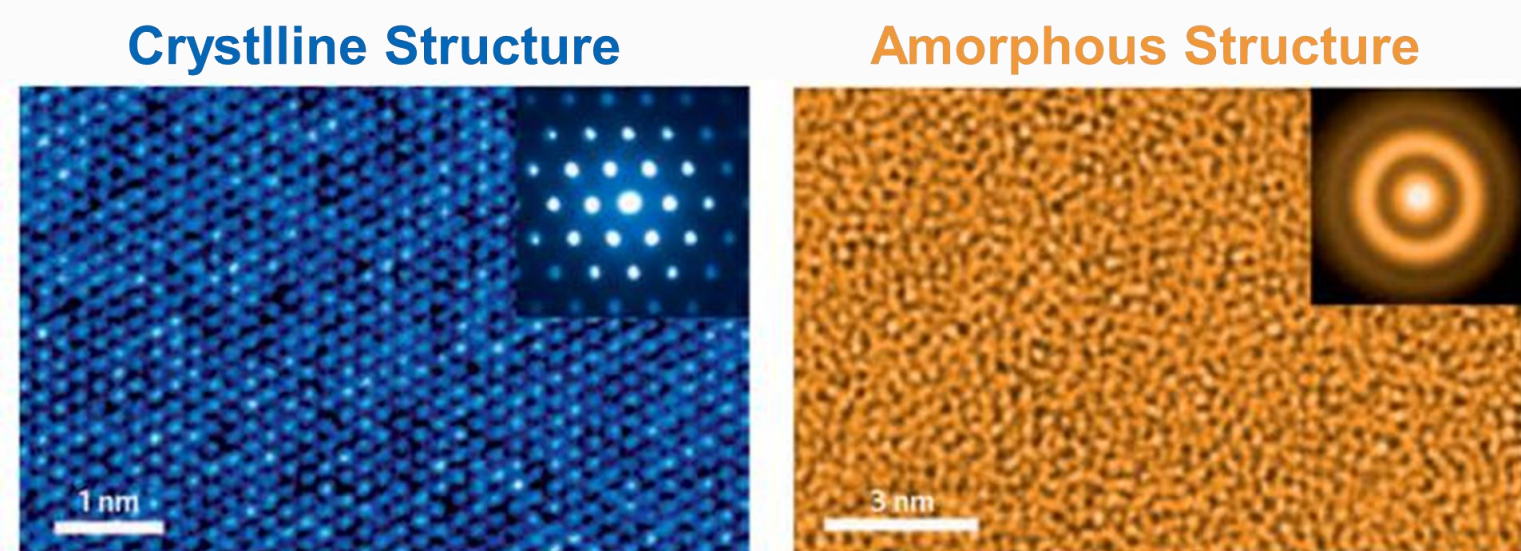
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What is Bulk Metallic Glasses (BMGs)?

Bulk metallic glasses (BMGs) are a type of metallic alloy that exhibit a disordered atomic structure, similar to that of glass, rather than the crystalline structure typically found in metals.

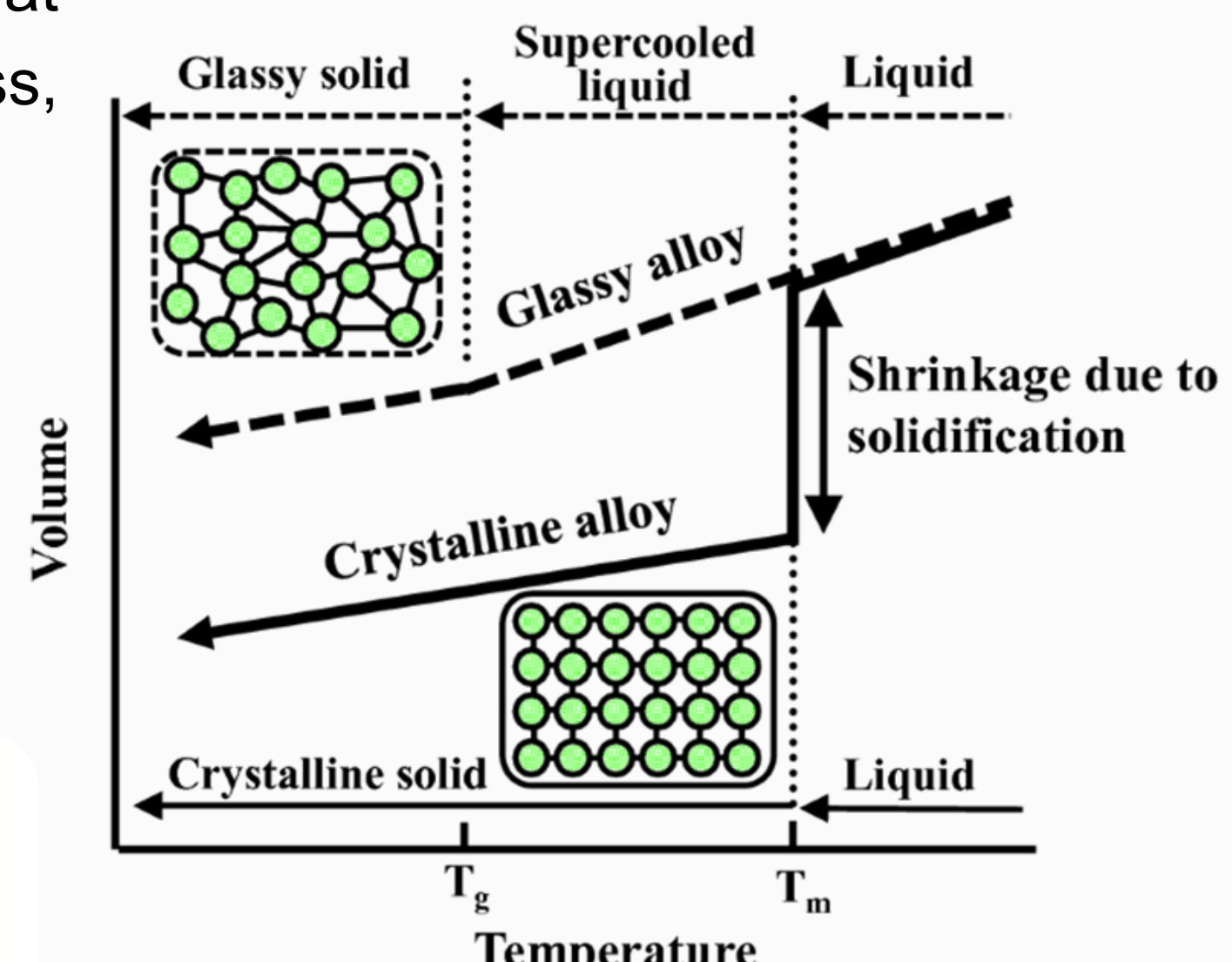


Chemical Properties

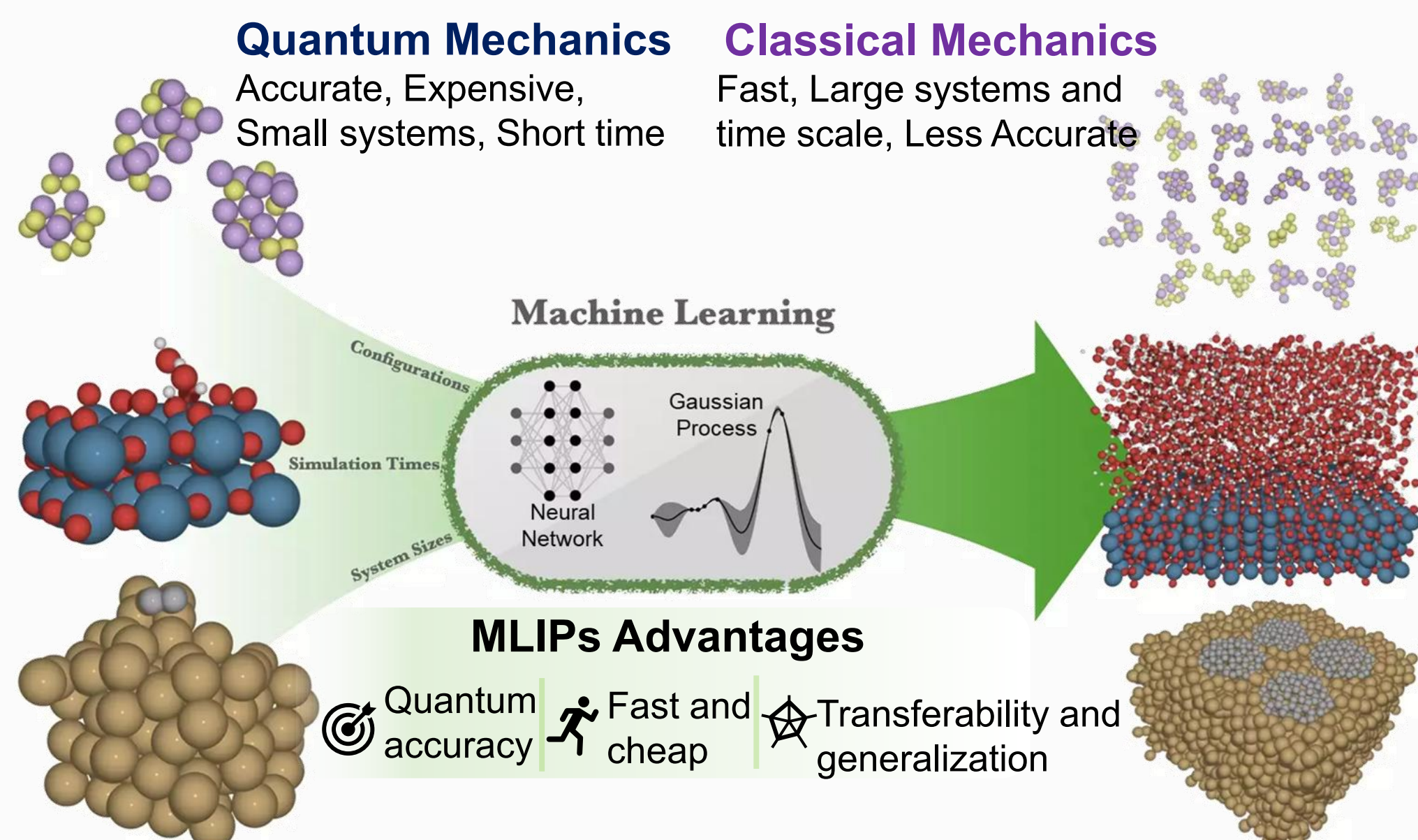
- ✓ Corrosion Resistance
- ✓ Potential as an Electro-catalyst
- ✓ Energy Storage

Mechanical Properties

- ✓ High Yield Strength
- ✓ High Tensile Strength
- ✓ Significant Elastic Strain



Why Machine Learning Interatomic Potentials (MLIPs)?



Why Does the Surface Need Special Treatment?

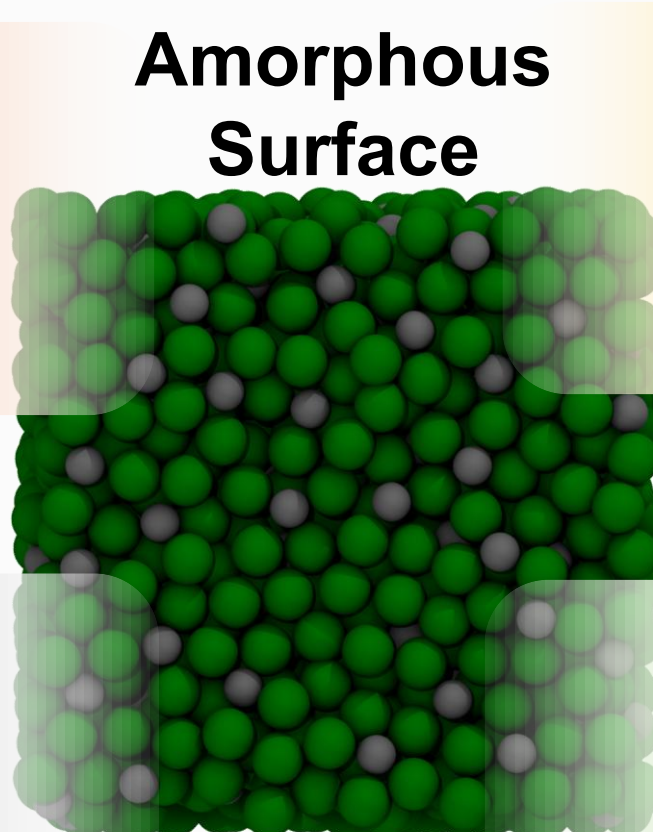
Bulk-trained models are not enough for surfaces:

Surface atoms have reduced coordination, broken symmetry, and distinct relaxations.

There is no single well-defined surface:

Unlike crystals, amorphous materials contain a broad distribution of local atomic motifs and terminations.

Surface-Aware Machine Learning Interatomic Potential (SA-MLIP) can capture undercoordinated and disordered surface sites that govern the interfacial chemistry



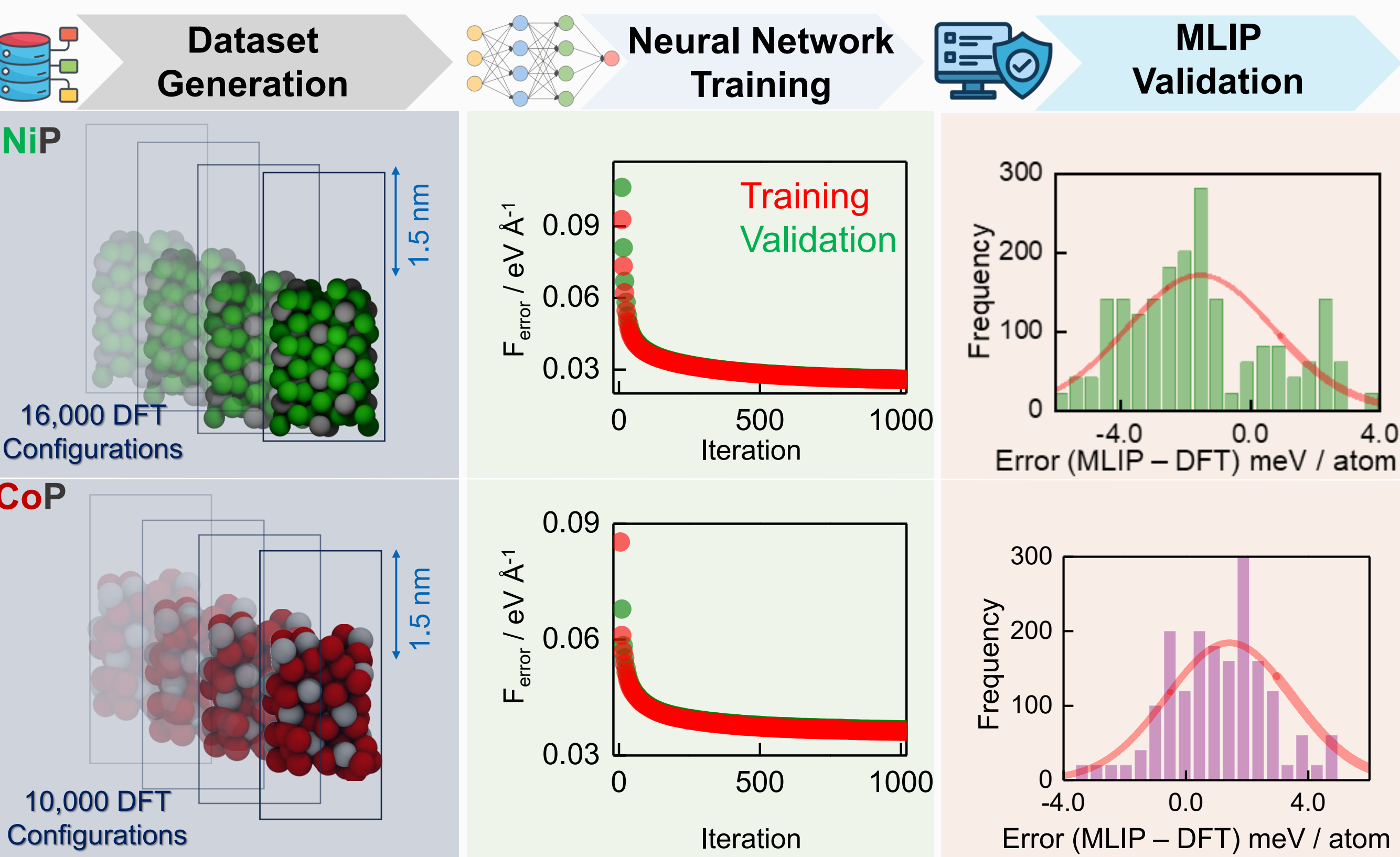
Catalysis is surface-dominated-

accurate prediction of surface energetics and exposure is essential for realistic catalyst modeling.

Surface reactivity depends on local environments:

In amorphous systems, small changes in coordination and local chemistry can create very different active sites.

How to Develop SA-MLIPs ?



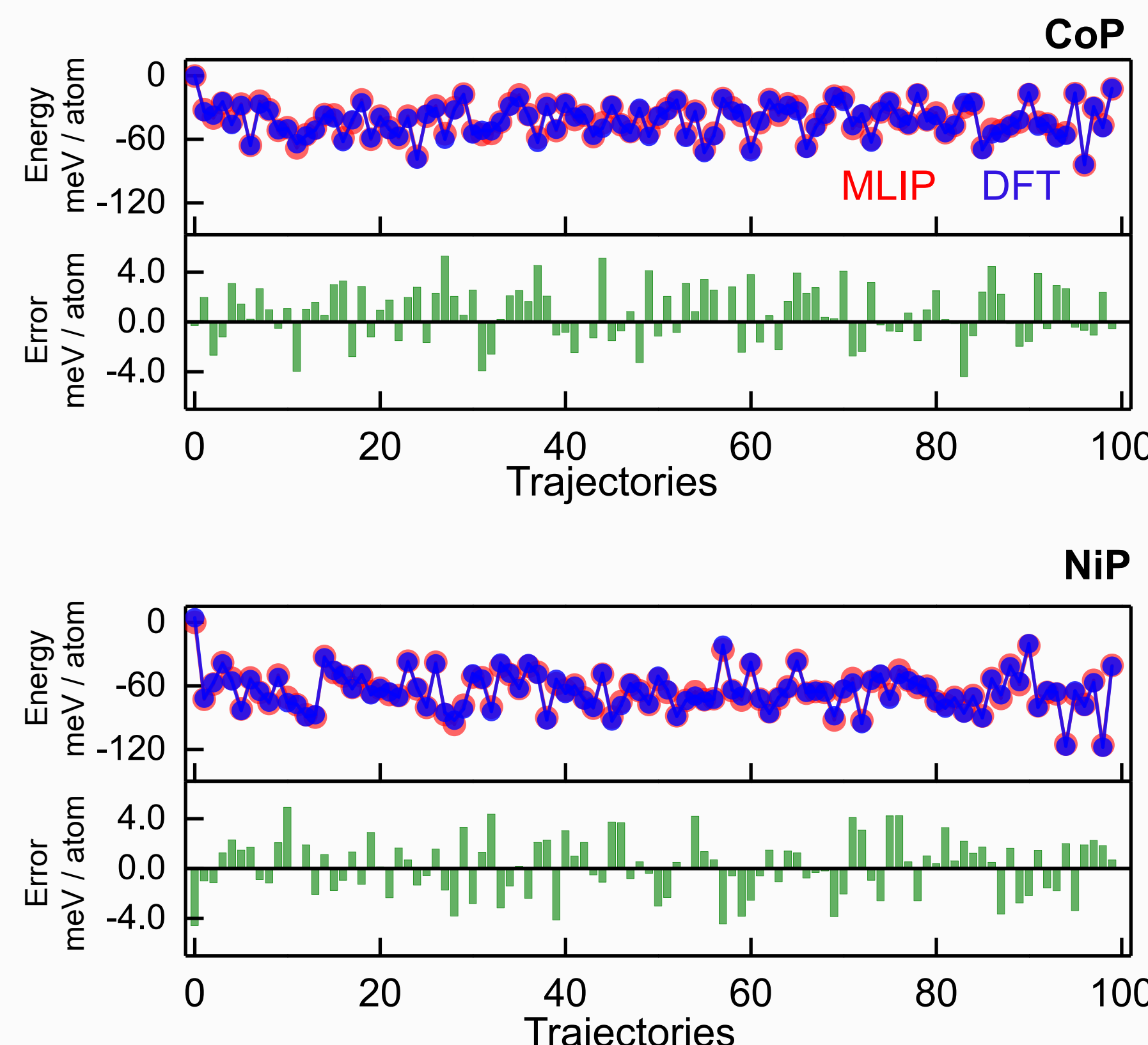
DFT calculations

- ✓ **Quantum ESPRESSO** with a plane-wave pseudopotential approach
- ✓ **Cutoffs:** 45 Ry for wave functions and 360 Ry for charge density
- ✓ **Smearing:** Gaussian, 0.01 Ry
- ✓ **SCF convergence:** 10⁻⁸ Ry, with local-TF mixing and Davidson diagonalization

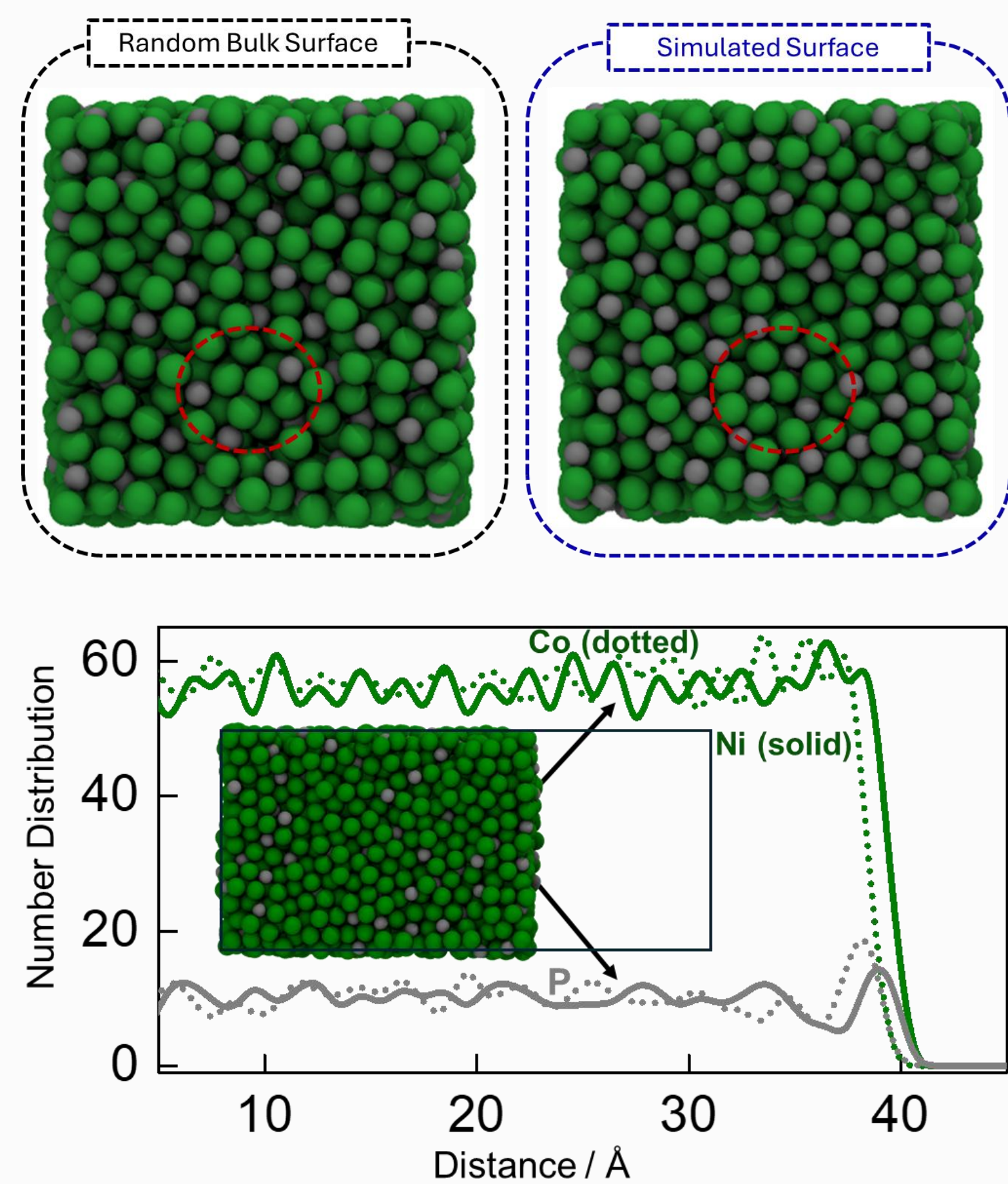
SA – MLIP Training Setups

- ✓ **Model:** Local equivariant **Allegro** neural network potential
- ✓ **Cutoff:** 5.0 Å, with automatic neighbor normalization
- ✓ **Descriptors:** Trainable **Bessel basis** + polynomial cutoff (**p = 5**)
- ✓ **Symmetry:** $l_{\max} = 2$, full **O(3)** parity
- ✓ **Nonlinearity:** SiLU with latent **ResNet** connections

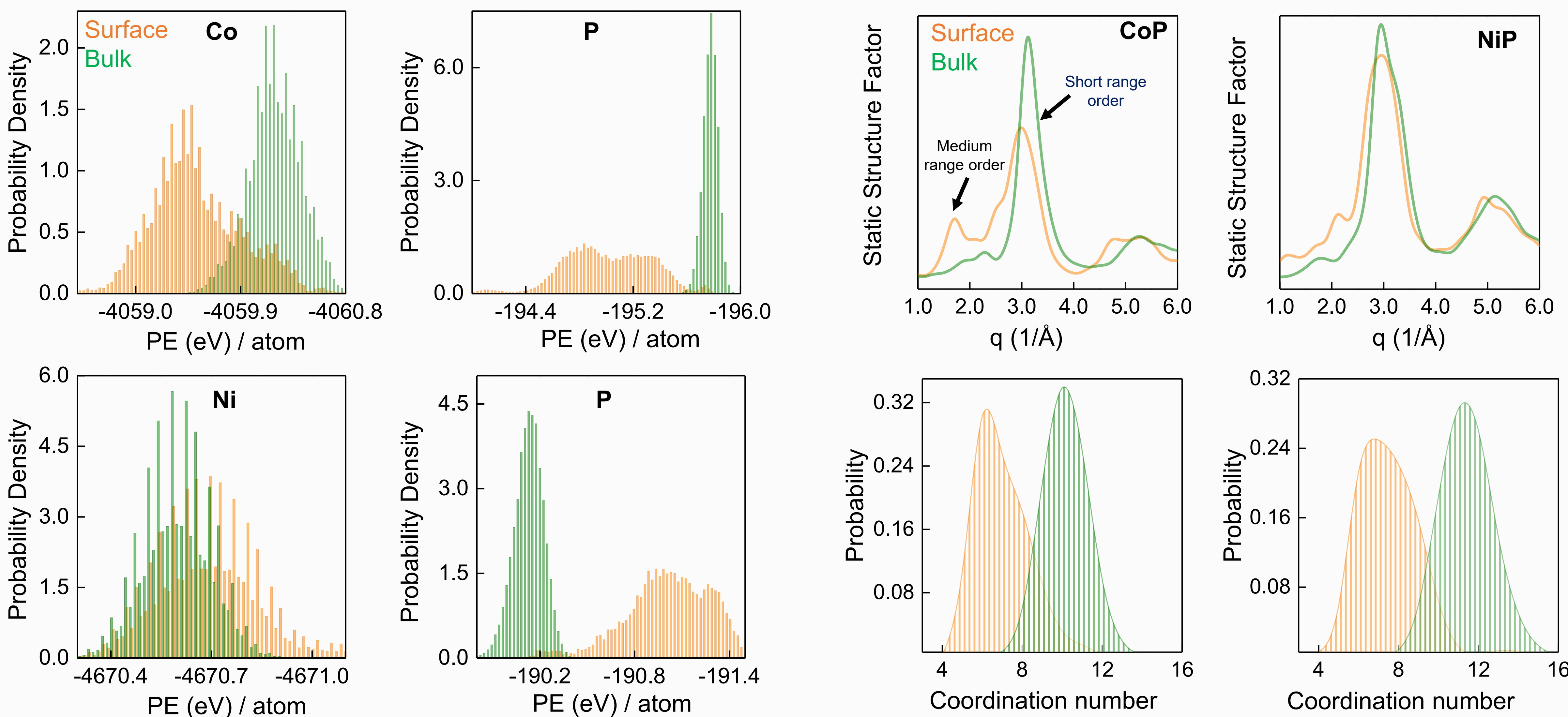
How Accurate Is the Potential?



SA-MLIPs achieve near-DFT accuracy, with errors below 3 meV / atom.



How Does the Surface Differ from the Bulk?



The potential energy (PE) changes significantly at the surface relative to the bulk.

Compared with the bulk, the surface shows a shifted local structural distribution, reflecting a distinct atomic environment.

What Are the Key Takeaways?

- ✓ Surface physics cannot be inferred from bulk alone in amorphous metal–phosphorus glasses.
- ✓ Our surface-specific MLIP bridges this gap by describing bulk and surface environments within one unified framework.
- ✓ The surface undergoes pronounced chemical and structural reorganization relative to the bulk.
- ✓ CoP and NiP display fundamentally different surface stabilization behavior.
- ✓ Metal–P short-range ordering emerges as a key signature linking local chemistry to surface energetics.
- ✓ This framework opens a scalable path for surface-sensitive discovery in catalytic bulk metallic glasses.

Where Can You Learn More?

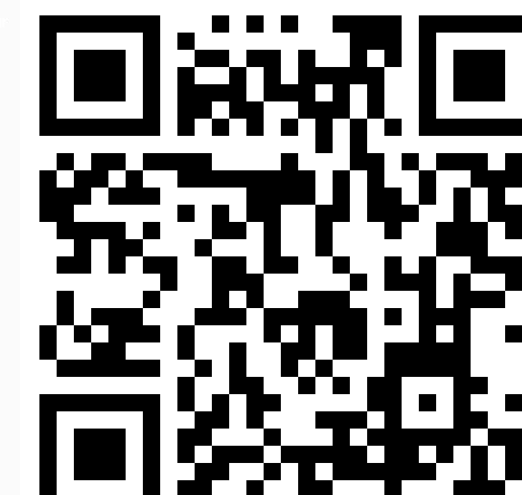
1. Hirata *et al.* Geometric Frustration of Icosahedron in Metallic Glasses. *Science* 341, 376–379 (2013).
2. Sheng, H., *et al.* Atomic packing and short-to-medium-range order in metallic glasses. *Nature* 439, 419–425 (2006)
3. Friederich, P. *et al.* Machine-learned potentials for next-generation matter simulations. *Nat. Mater.* 20, 750–761 (2021)
4. Khan, M. S. *et al.* Understanding Structure-Composition-Property Relationships of Ni-P Bulk Metallic Glasses. *J. Phys. Chem. C* 129, 19065–19073 (2025)
5. Yin, J., *et al.* SurFF: a foundation model for surface exposure and morphology across intermetallic crystals. *Nat. Comput Sci* 5, 782–792 (2025)
6. Khan, M. S. *et al.* Critical Composition Window for Co–P Bulk Metallic Glasses Revealed by Machine-Learned Interatomic Potentials. *J. Phys. Chem. C* 129, 22138–22146 (2025)
7. Musaelian, A. *et al.* Learning local equivariant representations for large-scale atomistic dynamics. *Nat Commun* 14, 579 (2023)

Who Supported This Work?



BOISE STATE UNIVERSITY

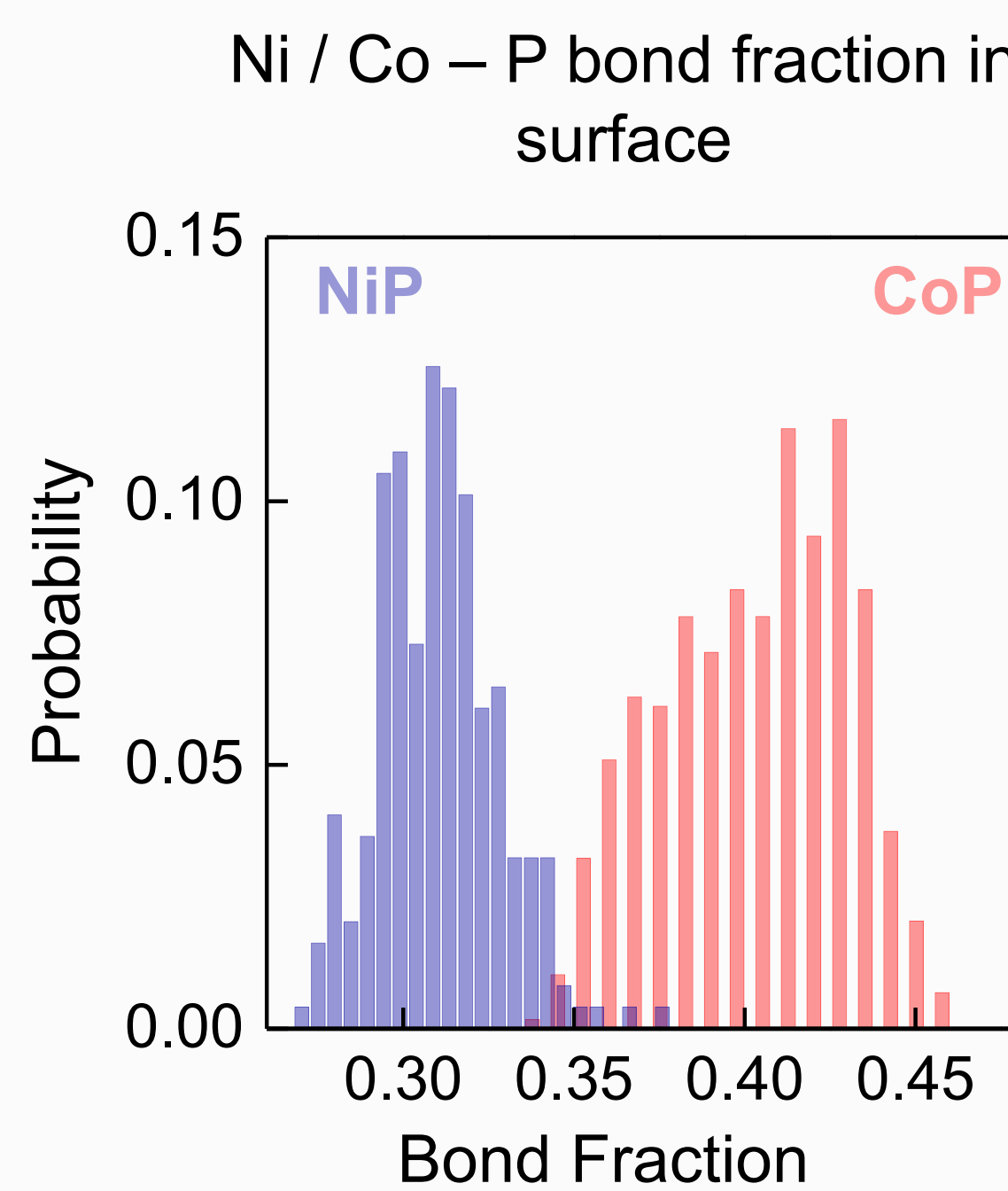
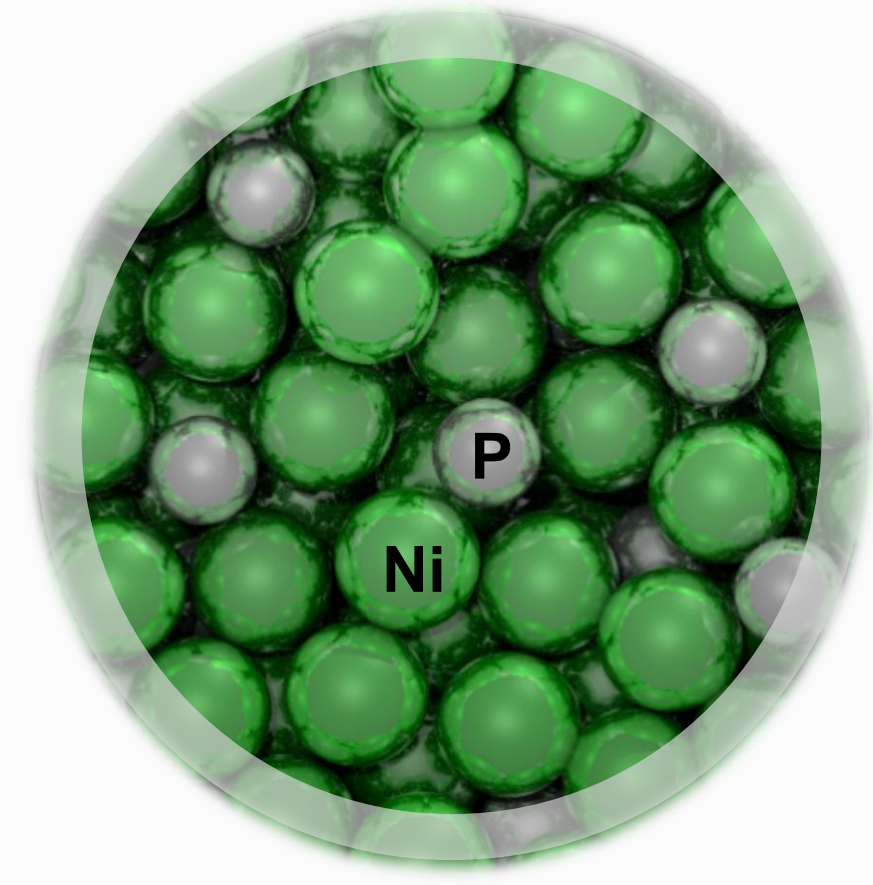
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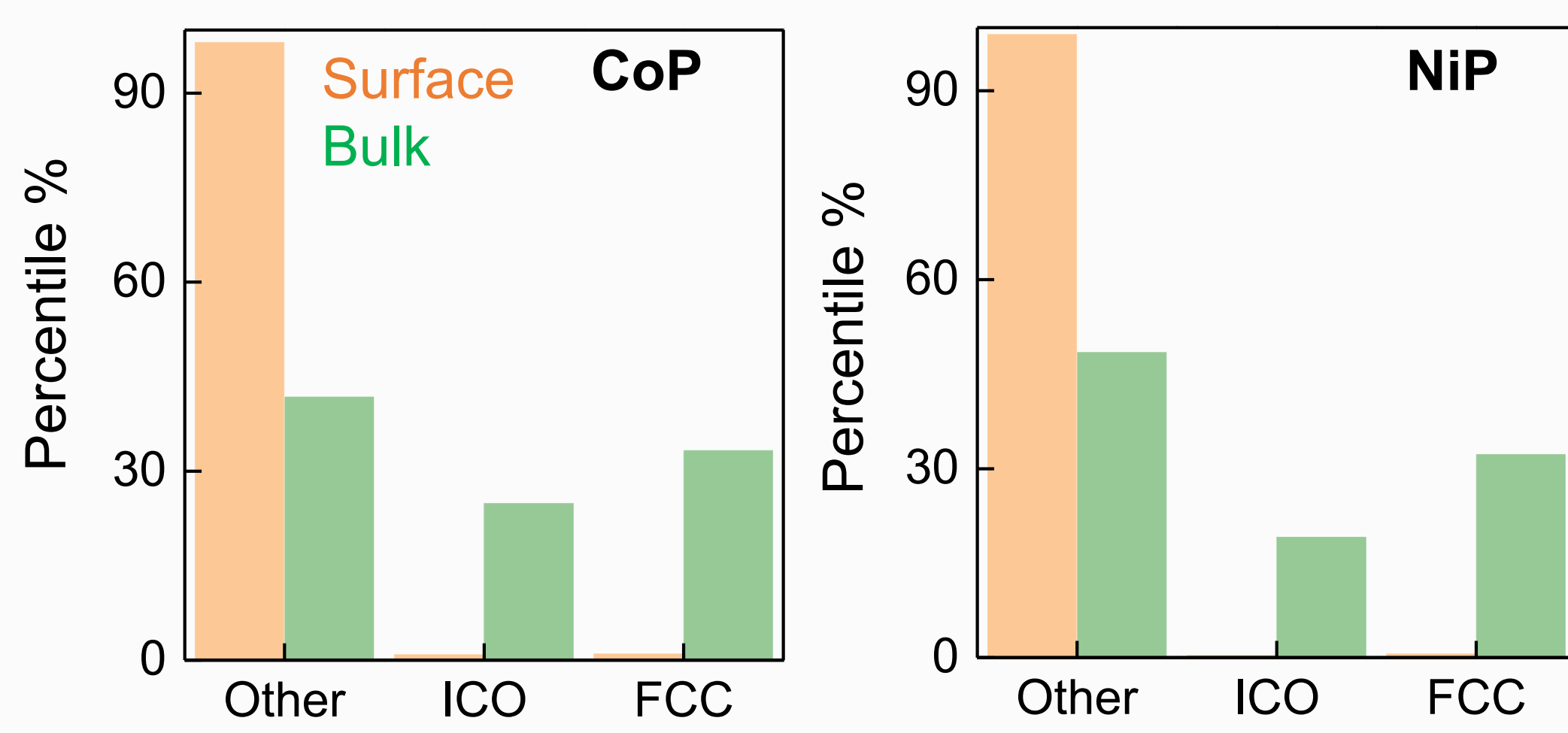
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Simulated surface atomic arrangement of NiP



Polyhedral template matching analysis



Warren – Cowley – like short range ordering

